

Ethics of AI #2

ECE Paris – Ing5 Data&IA



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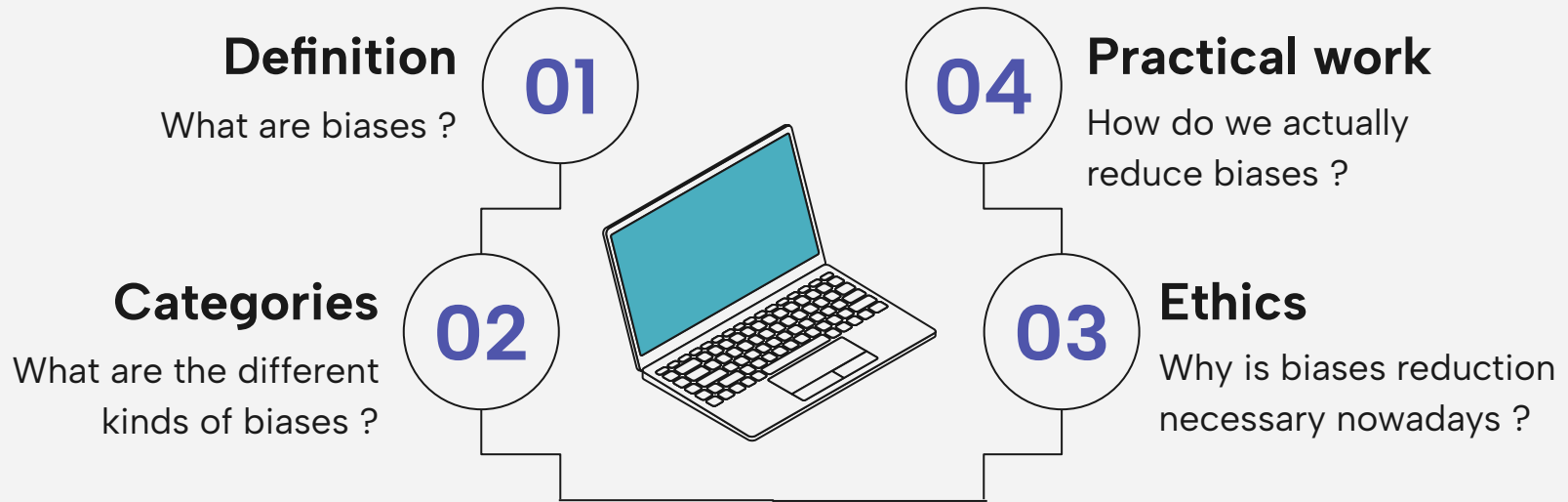
Overview of the course

5 two-hours classes, including two+ practical works and one graded project

1. **Culture of AI Ethics** : lecture and group presentations
2. **Biases** : lecture and practical work 👉
3. **Fine-tuning** : lecture and practical work
4. **XAI** : *lecture and practical work* / **Project** : *start of group work ?*
5. **Project** : graded group presentations

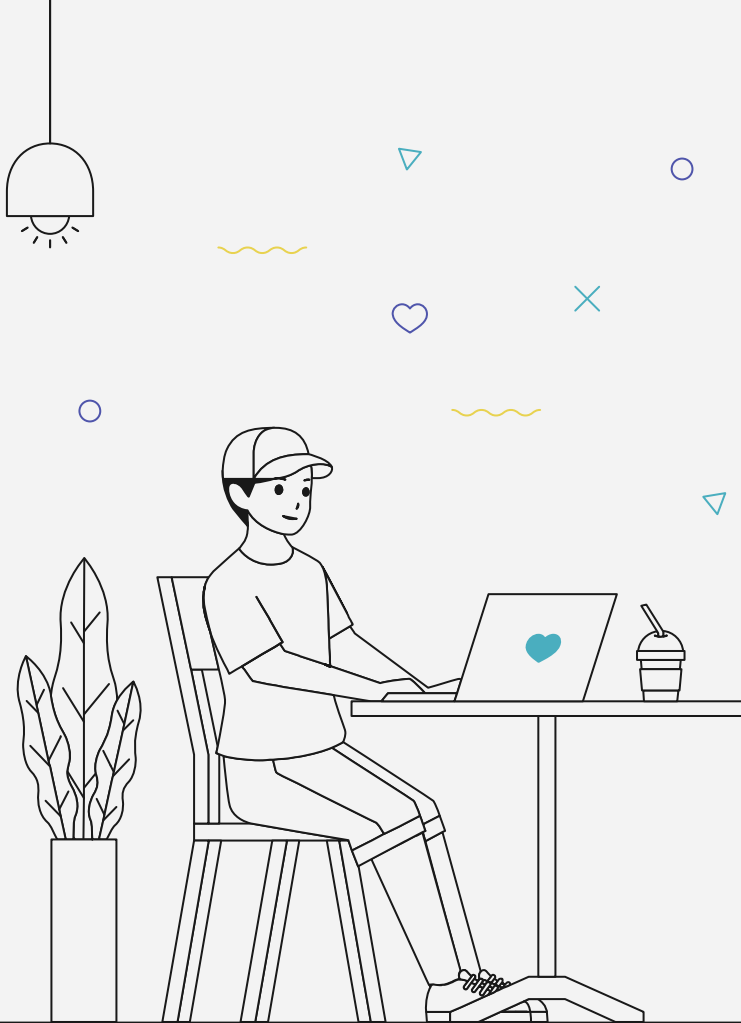


Biases : contents



01 Definition

What are biases ?





1. Definition



Systematic error in an system that produces **unfair, inaccurate** or **discriminatory** results against certain **individuals** or **groups**.

The fundamental problem: bias is a **systemic** and **reproducible injustice**, often inherited from **data** or **human choices**.

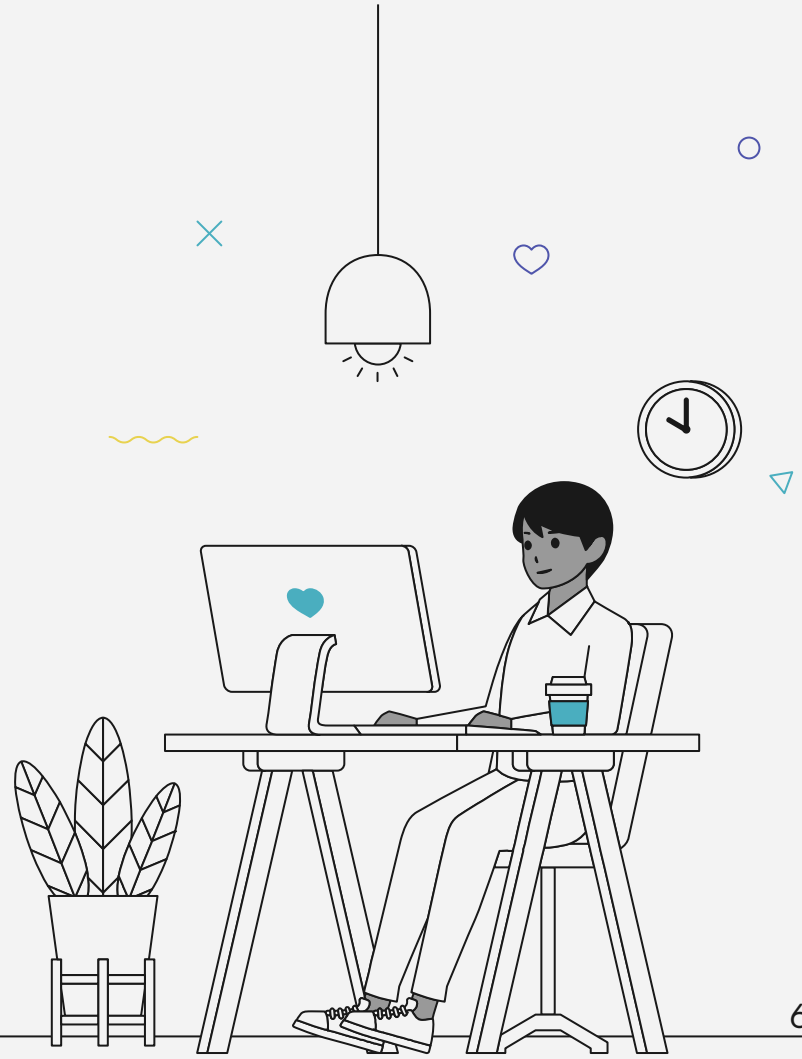
Biases are an inevitable **reflection** of **human nature**, given that AI is an **artificial** discipline, and then need to be **reduced** as much as possible.



02

Categories

What are the different kinds of biases ?





2.1. Data bias

Bias originating from the **dataset** used for **training**. The model learns **stereotypes** and **imbalances** present in the data.

Examples :

A recruitment system trained on resumes of **predominantly male** employees will **disadvantage female** applicants.

A visual recognition model that performs **well on light skin** but **fails on dark skin**, due to an imbalance in the **training images**.





2.2. Algorithmic bias



Biases introduced by the **choice** of algorithm, its **parameters** or by the very definition of the problem to be optimized. Example :

Scenario: A resume screening tool used by a large company.

The bias: The algorithm is trained to identify “good profiles” by **looking for similarities** with employees who are **currently** performing well in the **company**.

Impact: The system favors **cloning**. A brilliant candidate with an **atypical background** (career change, lesser-known school, experience abroad) will be **poorly rated** because they do not “resemble” the standard model.





2.3. Evaluation bias



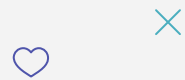
Bias that occurs when the **metrics** used to evaluate model performance are **inadequate** or do **not capture** notions of **fairness**.

Examples :

Measuring the “overall accuracy” of a lending model when it systematically **denies** loans to a **particular community**. Overall accuracy may remain high.

Failing to **specifically** test performance on **demographic subgroups**.





2.4. Interaction/Usage Bias



Bias emerging from the **interaction** between **users** and the AI **system** after its deployment.

Examples :

Video **recommendations** that create “echo chambers” and **polarize opinion**.

A **gendered** voice assistant (e.g., default female voice for assistants) that **reinforces** gender **stereotypes**.



03 Ethics

Why is biases reduction necessary today ?





3.1. Ethical implications of biases



Data : **crystallize** and **automate** past **discrimination**, giving it an **appearance** of technical **neutrality**.

Algorithmic : the decision **appears** to be “**mathematical**” and “objective,” **masking** the human design **choices** that are themselves laden with **values** and **judgments**.

Evaluate : **relying** on misleading metrics can give a false sense of **confidence** and lead to the **deployment** of a deeply **unfair** system.

Interaction/Usage : AI is not a passive tool; it **changes** users' **behaviors** and **perceptions** that can **amplify initial** biases.





3.2. Towards Fairer AI

Detecting and reducing bias is a **process**, not a simple checkbox.

Possible Solutions

Upstream: Data audits, diversification of development teams.

During: Use of fairness metrics, testing on subgroups.

Downstream: Transparency, complaint systems, continuous monitoring after deployment.





Conclusion :

- Understanding biases means recognizing that AI is a product of society.
- Making it fair is therefore a collective responsibility, as much technical as it is social and political.



04

Practical work

How do we actually reduce biases ?





4. Practical work

Use case : Automated recruitment system

Goal : Understanding, detecting, and correcting biases in data and algorithms

Tool : Google Colaboratory highly preferred





**Thanks for your
attention.**

